

Program10: Demonstrate Kubernetes Dashboard and CLI Cluster Monitoring

Project Structure

DEVOPSLAB

--program10

--deployment.yaml

--service.yaml

Step 1: Initiate minikube allocating specific CPUs and memory

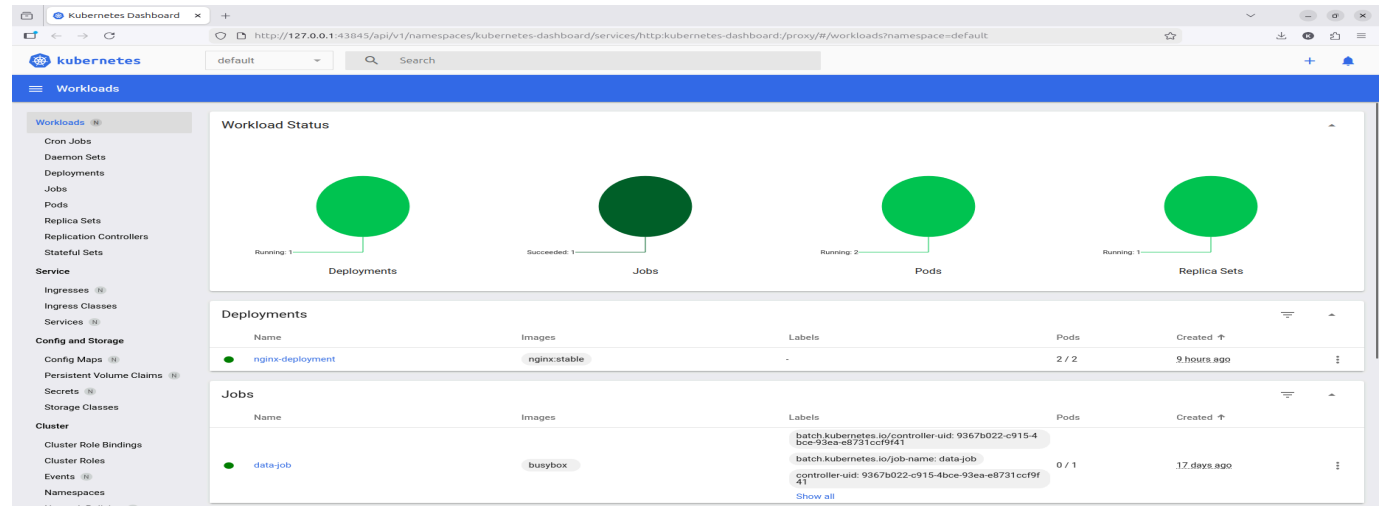
Step 2: Enable Kubernetes Dashboard

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program10
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~$ cd DEVOPSLAB
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ mkdir program10
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd program10
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ minikube start --cpus=2 memory=4096
🌟 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🐳 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ... ■ Using image docker.io/kubernetes/dashboard:v2.7.0
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔧 Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard

! /usr/local/bin/kubectl is version 1.28.4, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ minikube dashboard
🌟 Verifying dashboard health ...
🚀 Launching proxy ...
🌟 Verifying proxy health ...
👉 Opening http://127.0.0.1:43845/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 10:04:23.247: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
[92772, Main Thread] WARNING: GTK+ module /snap/firefox/7477/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot
```



Step 3: Create a Deployment file

```
1rv24mc054_khushijs@khushi-CF63-Thin-105CXR: ~/DEVOPSLAB/program10
GNU nano 7.2 deployment.yaml *
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Help Write Out Where Is Cut Execute Location Undo Set Mark
Exit Read File Replace Paste Justify Go To Line Redo Copy

Step 4: Create a Service file

```
1rv24mc054_khushijs@khushi-CF63-Thin-105CXR: ~/DEVOPSLAB/program10
GNU nano 7.2 service.yaml *
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

Help Write Out Where Is Cut Execute Location Undo Set Mark
Exit Read File Replace Paste Justify Go To Line Redo Copy

Step 5: Apply Resources

Step 6: Monitor via CLI

Step 7: Check Nodes

Step 8: Check services

Step 9: Check pod logs

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program10
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~$ cd DEVOPSLAB
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd program10
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ nano deployment.yaml
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ nano service.yaml
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl apply -f service.yaml
service/nginx-service unchanged
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7d584c4448-d4d24   1/1     Running   0           27s
nginx-deployment-7d584c4448-g94l4   1/1     Running   0           22s
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  18d   v1.34.0
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1     <none>        443/TCP          9h
nginx-service  NodePort    10.100.147.38 <none>        80:32103/TCP     9h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl logs pod nginx-deployment-7d584c4448-d4d24
Error from server (NotFound): pods "pod" not found
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl logs nginx-deployment-7d584c4448-d4d24
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:38:31 [notice] 1#1: using the "epoll" event method
```

Step 10: Describe pods

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program10
2025/12/19 04:38:31 [notice] 1#1: start worker process 36
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ kubectl describe pod nginx-deployment-7d584c4448-d4d24
Name:                                nginx-deployment-7d584c4448-d4d24
Namespace:                           default
Priority:                             0
Service Account:                      default
Node:                                minikube/192.168.49.2
Start Time:                           Fri, 19 Dec 2025 10:08:27 +0530
Labels:                               app=nginx
                                      pod-template-hash=7d584c4448
Annotations:                          <none>
Status:                               Running
IP:                                   10.244.1.160
IPs:
  IP:                                 10.244.1.160
Controlled By:                        ReplicaSet/nginx-deployment-7d584c4448
Containers:
  nginx:
    Container ID:  docker://2220a96121f190f302528f6f68fdae40b20268f538d1e55c99e84338ae1b2cf4
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 19 Dec 2025 10:08:30 +0530
      Ready:      True
      Restart Count: 0
      Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-95hdh (ro)
Conditions:
  Type          Status
```

Step 11: Access the application via minikube and viewing the same in the web browser

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program10
Normal Scheduled 117s default-scheduler Successfully assigned default/nginx-deployment-7d584c4448-d4d24 to minikube
Normal Pulled 116s kubelet Container image "nginx:latest" already present on machine
Normal Created 114s kubelet Created container: nginx
Normal Started 114s kubelet Started container nginx
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32103

🐞 Opening service default/nginx-service in default browser...

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program10$ Gtk-Message: 10:10:42.996: Not loading module "atk-bridge": The
functionality is provided by GTK natively. Please try to not load it.
[95419, Main Thread] WARNING: GTK+ module /snap/firefox/7477/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so canno
t be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/par
ts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:95419): Gtk-WARNING **: 10:10:43.055: GTK+ module /snap/firefox/7477/gnome-platform/usr/lib/gtk-2.0/modules/libcan
berra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 10:10:43.055: Failed to load module "canberra-gtk-module"
[95419, Main Thread] WARNING: GTK+ module /snap/firefox/7477/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so canno
t be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/par
ts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:95419): Gtk-WARNING **: 10:10:43.056: GTK+ module /snap/firefox/7477/gnome-platform/usr/lib/gtk-2.0/modules/libcan
berra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 10:10:43.056: Failed to load module "canberra-gtk-module"
```



Step 12: Open the minikube dashboard and monitor the status of the all pods

Workload Status

Running: 1 Deployments Succeeded: 1 Jobs Running: 2 Pods Running: 2 Replica Sets

Name	Images	Labels	Pods	Created ↑
nginx-deployment	nginx:latest	app: nginx	2 / 2	9 hours ago

Name	Images	Labels	Pods	Created ↑
data-job	busybox	batch.kubernetes.io/controller-uid: 9367b022-c915-4bce-93ea-e8731ccf9f41 batch.kubernetes.io/job-name: data-job controller-uid: 9367b022-c915-4bce-93ea-e8731ccf9f41	0 / 1	17 days ago